Week 1 Lab Report

Setup of a Cybersecurity Testing Lab using VirtualBox and Kali Linux

Networkwalks Academy: Week 1 Project Module 1 (WK1-PM1)

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1. Objective

The aim of this task was to build a small, isolated cybersecurity lab on a Windows 11 host. Oracle VirtualBox serves as the hypervisor, and a Kali Linux virtual machine acts as the attacking machine. The guest is placed on a custom NAT Network in the 10.0.0.0/24 subnet, configured with clipboard and drag-and-drop support, given a shared folder with the host, and confirmed to have full internet access. A snapshot of the finished configuration was taken so the lab can be restored.

2. Lab Environment

Component	Details
Host operating system	Windows 11
Hypervisor	Oracle VirtualBox Manager
Guest operating system	Kali Linux 2026.3 (kali-linux-2026.3-virtualbox-amd64)
Guest base memory	2048 MB
Network	NAT Network "NatNetwork", IPv4 prefix 10.0.0.0/24, DHCP enabled, IPv6 disabled
Kali IP address	10.0.0.2/24 on interface eth0
Snapshot	Fresh Snapshot, taken 27 October 2026

3. Procedure and Evidence

3.1 Kali Linux running in VirtualBox

The Kali Linux 2026.3 guest was started from the VirtualBox window. The desktop loads normally, which confirms the image imported correctly and boots without errors.

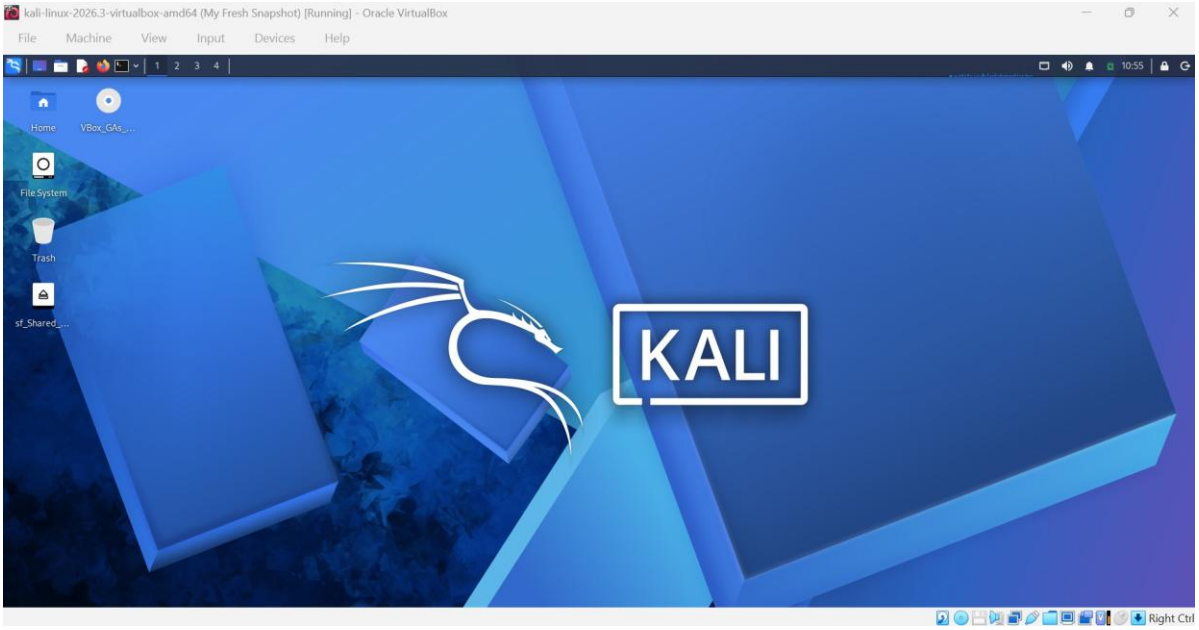


Figure 1. Kali Linux desktop running inside VirtualBox (snapshot: Fresh Snapshot).

3.2 NAT Network configuration

A NAT Network named NatNetwork was created under File, then Tools, then Network. Its IPv4 prefix is set to 10.0.0.0/24 with DHCP enabled, and IPv6 is left disabled. This satisfies the requirement to use NATNetwork on the 10.0.0.0/24 subnet.

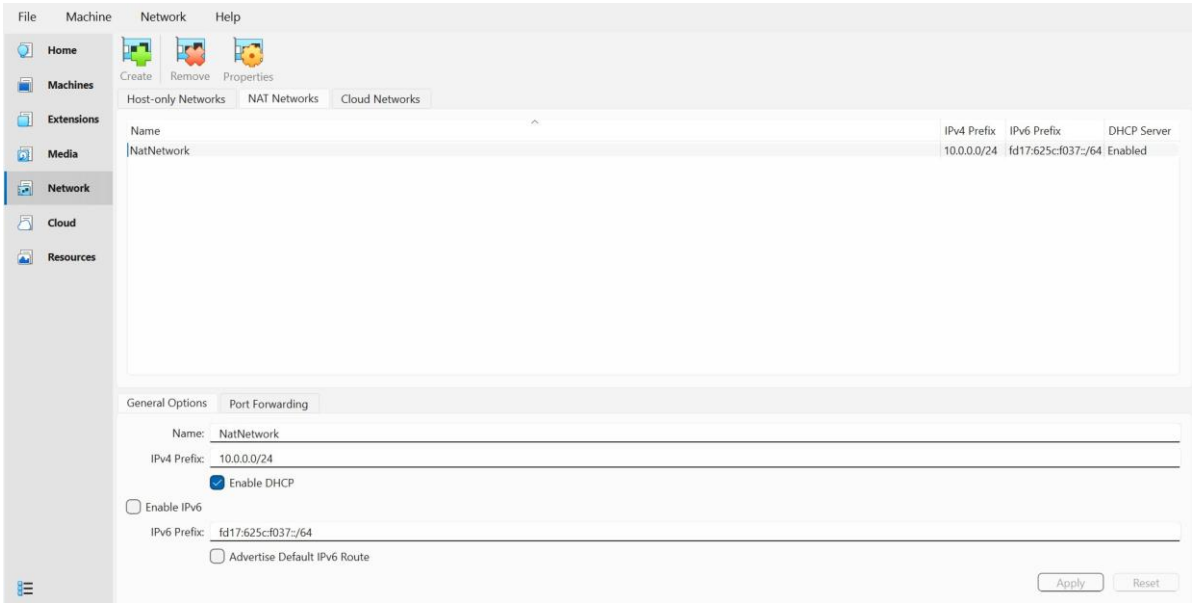


Figure 2. Oracle VirtualBox Network Manager showing NatNetwork (10.0.0.0/24, DHCP enabled).

3.3 Static IP address on Kali

The address was checked from inside the guest with the command ip -4 addr show eth0. The interface reports inet 10.0.0.2/24, which matches the required Kali IP address and is inside the NAT Network range.

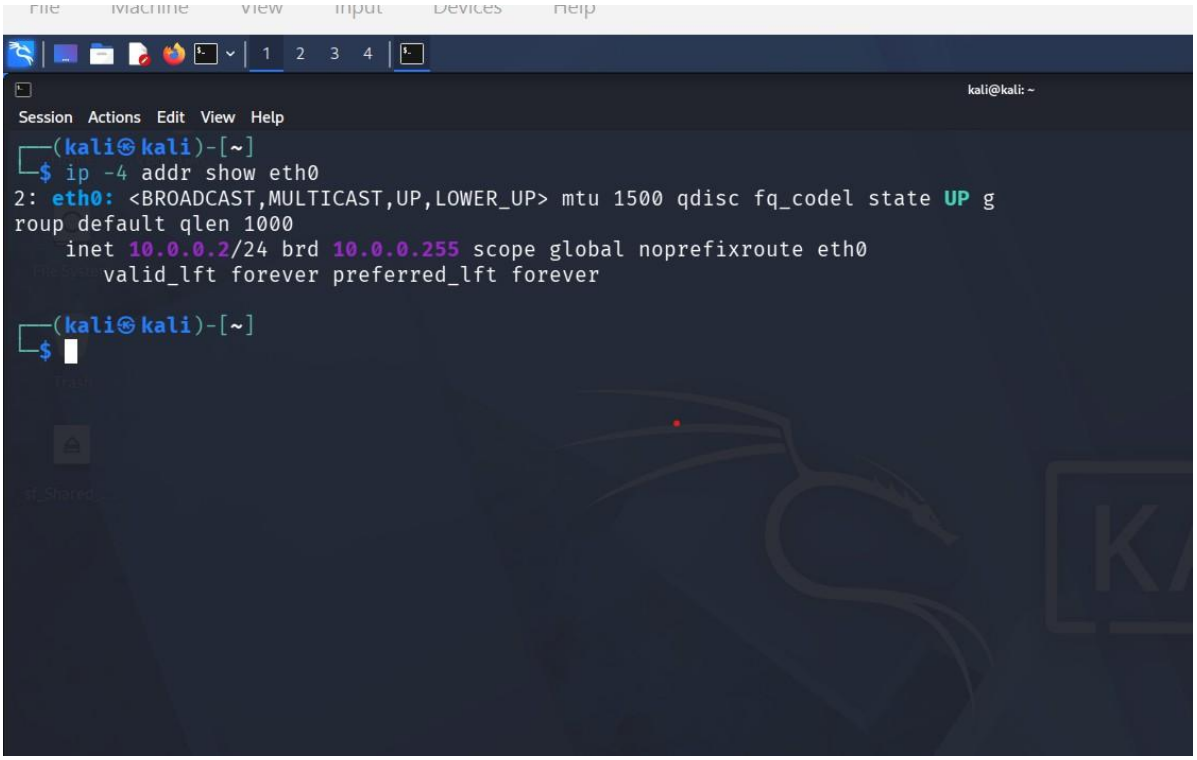


Figure 3. Output of ip -4 addr show eth0 confirming 10.0.0.2/24 on eth0.

3.4 Internet connectivity

Outbound connectivity was tested with ping -c 4 google.com. All four ICMP echo requests received replies, with 0% packet loss and an average round-trip time of about 274 ms. This confirms that Kali has full internet access through the NAT Network.

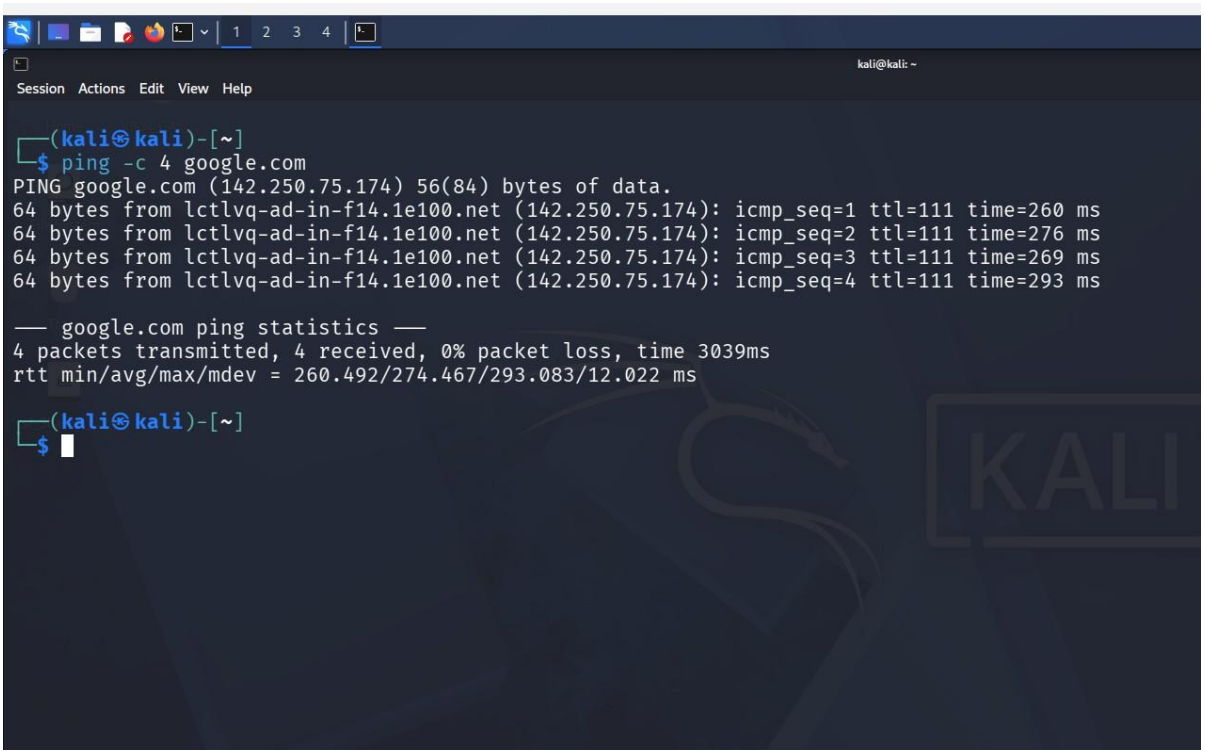


Figure 4. ping results for google.com: 4 transmitted, 4 received, 0% packet loss.

3.5 Shared clipboard and drag-and-drop

In the VM's General settings under the Features tab, both Shared Clipboard and Drag-and-Drop are set to Bidirectional. This allows text and files to move between the Windows 11 host and the Kali guest in both directions.

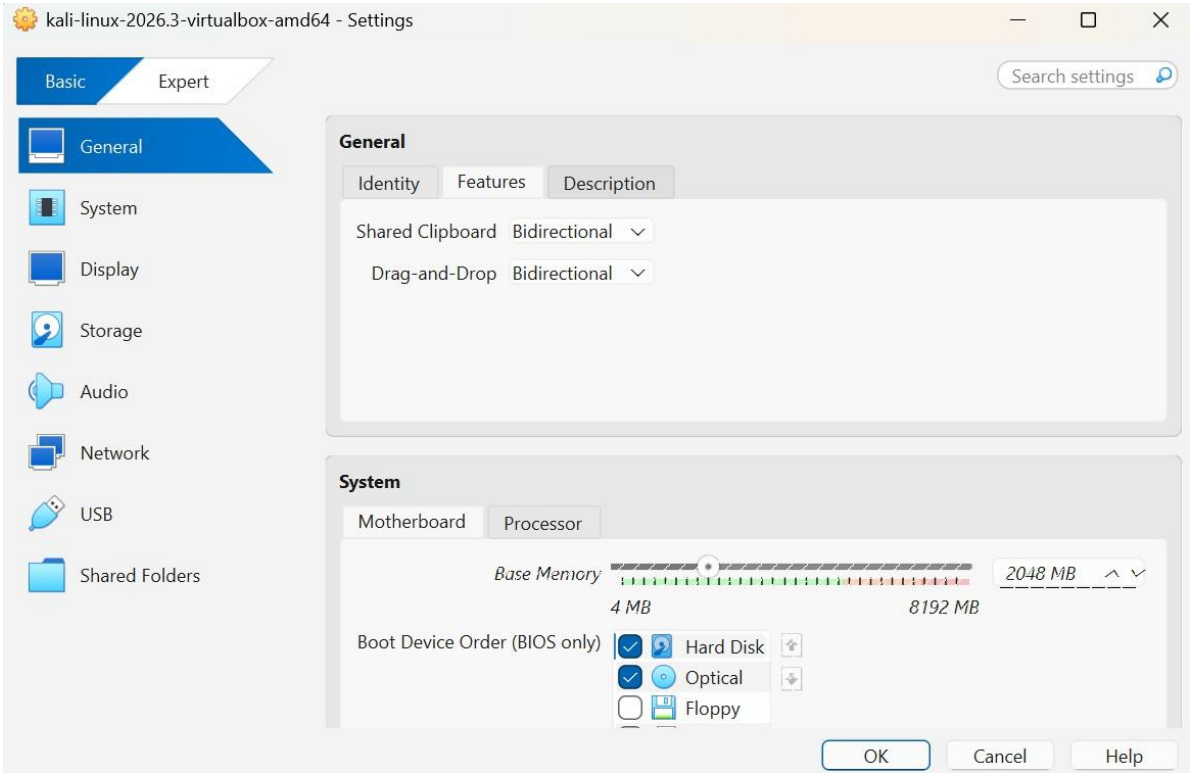


Figure 5. VM settings with Shared Clipboard and Drag-and-Drop set to Bidirectional.

3.6 Shared folder between host and guest

A shared folder was created in the VM's Shared Folders settings. The host path is E:\Shared Folder, exposed to the guest as Shared_Folder with Full access, Auto-mount enabled and Make Machine-permanent enabled. The brief asks for the host's Downloads folder to be shared; this lab uses E:\Shared Folder instead, which uses the same mechanism.

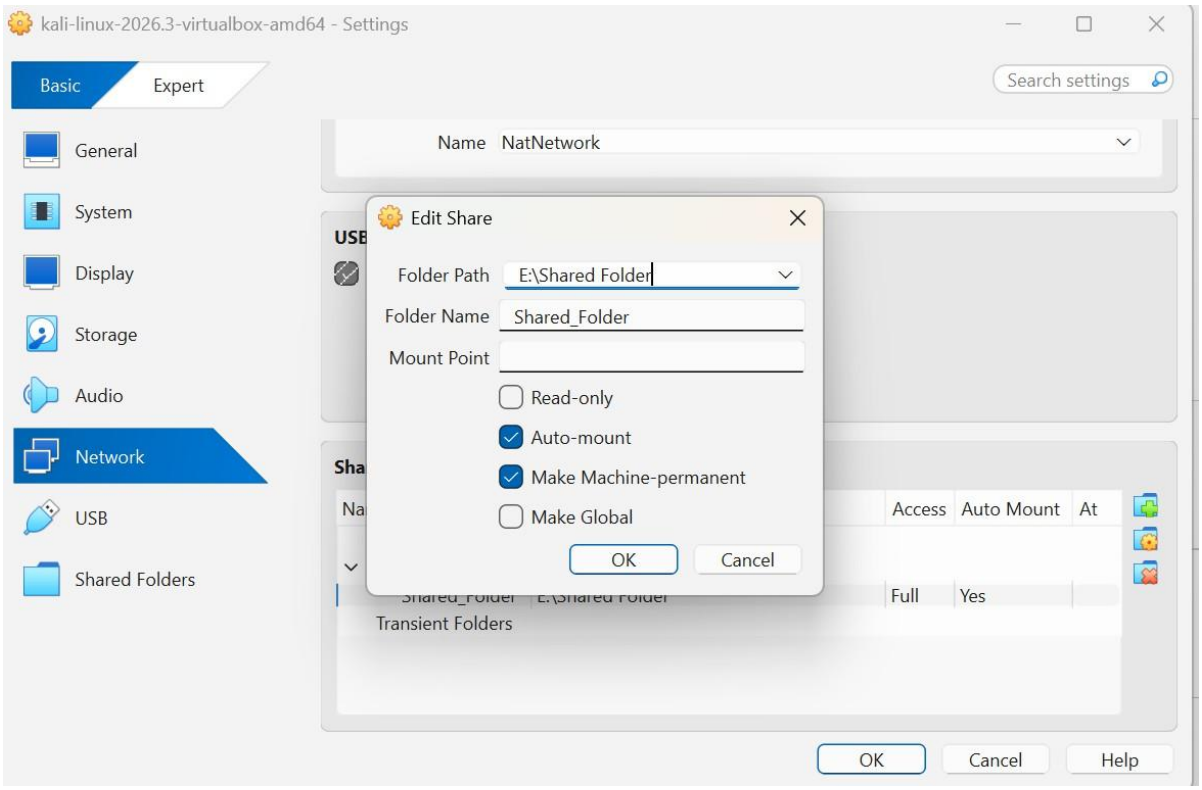


Figure 6. Edit Share dialog for Shared_Folder (host path E:\Shared Folder).

3.7 Snapshot of the configured VM

Once the network, clipboard, drag-and-drop, shared folder and IP configuration were confirmed, a snapshot named Fresh Snapshot was taken on 27 October 2026. The snapshot lets the lab be restored to this known-good state at any time. Its description records the features enabled, and it refers to the Kali IP address.

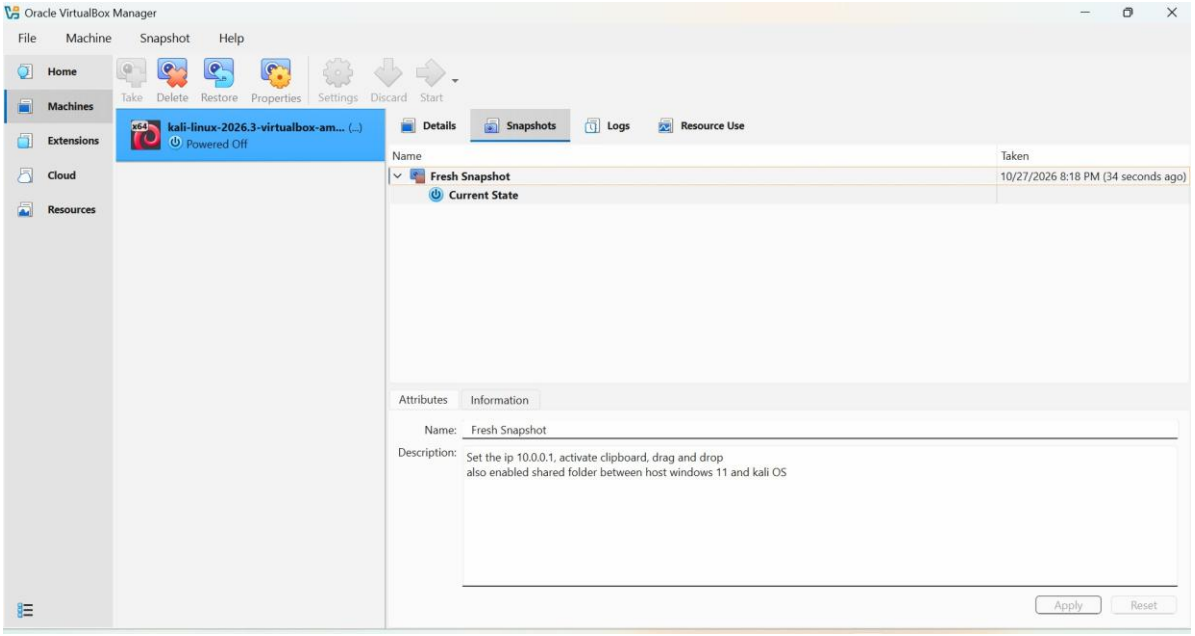


Figure 7. VirtualBox Manager showing the Fresh Snapshot and its description.

4. Requirements Checklist

Each requirement from the WK1-PM1 task was checked against the evidence above. The table gives the outcome of each check.

#	Requirement	Evidence	Result
1	Use VirtualBox as the base	Figures 1, 2, 7	Met
2	Set up Kali Linux as the attacking machine	Figure 1	Met
3	Network in subnet 10.0.0.0/24	Figure 2	Met
4	Use NATNetwork with 10.0.0.0/24	Figure 2	Met
5	Clipboard and drag-and-drop enabled	Figure 5	Met
6	Shared folder enabled from host	Figure 6	Met (path differs from brief)
7	Kali IP address 10.0.0.2/24	Figure 3	Met
8	Full internet access	Figure 4	Met

5. Observations and Notes

The shared folder deviates from the brief. The task specifies the host's Downloads folder, but this lab shares E:\Shared Folder. The sharing setup is otherwise identical, so the requirement is functionally met. Changing the host path to the Downloads folder is a single edit in the Shared Folders settings if the assessor requires it exactly.

The snapshot description refers to the IP address 10.0.0.1, while the verified address on eth0 is 10.0.0.2/24. 10.0.0.1 is the gateway address on this subnet. The description is a note only; the live configuration matches the task.

Internet access worked on the first test. The course troubleshooting guide notes that VirtualBox 7 with Kali 2026.1 or newer can show DAD-related connectivity delays. If that occurs, the nmcli commands in the README resolve it.

6. Conclusion

The cybersecurity testing lab is running as specified. Kali Linux operates as the attacking machine on the 10.0.0.0/24 NAT Network with the static address 10.0.0.2/24. It has full internet access, bidirectional clipboard and drag-and-drop, and a shared folder with the host. A snapshot preserves this configuration. The lab is ready for the Phase 2 tasks, which add Windows and Android virtual machines to the same network.